

# Hands-On-Learning: Exploring the CIA Triad on Endpoints

# Learning Objectives

By the end of this activity, learners will be able to:

- Map endpoint security risks to the CIA Triad.
  - Configure a basic Windows security setting to improve confidentiality, integrity, and availability.
  - Document their findings and explain their security choices.
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# Structure of the Activity

## **Title:** Applying the CIA Triad in Endpoint Security

### **Tool used:**

Windows 10/11 (VM or local machine) -> Professional/  
Enterprise Edition

Group Policy Editor (or Local Security Policy)

MS Word / Google Docs for documentation

# Scenario

You are the IT security analyst for a small company. Employees often use laptops to access sensitive files. Recently, there were concerns about unauthorized USB device usage, unpatched applications, and service downtime. You are tasked to map these risks to the CIA Triad and apply one technical control for each.

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# Objective

- Understand how real-world endpoint risks map to the CIA Triad.
- Apply and document one endpoint configuration control per CIA principle.



# Instructions for Learners

# Step 1: Map Risks to CIA Triad

List three risks: unauthorized USB usage, outdated software, and service downtime.

Identify whether each risk affects Confidentiality, Integrity, or Availability.

# Step 2: Apply a Control for Confidentiality

Open gpedit.msc (Group Policy Editor).

Navigate to: Computer Configuration → Administrative Templates → System → Removable Storage Access.

**Block access to USB storage devices.**

## Step 3: Apply a Control for Integrity

Open Windows Update Settings.

Enable **automatic updates** to ensure system integrity through patches.

## Step 4: Apply a Control for Availability

Open Services.msc.

Select a critical service (example, Windows Update, Security Center).

Set it to **Automatic (Delayed Start)** to ensure it always runs.

# Step 5: Document Your Work

Take **screenshots** of each change.

Explain in a Word/Google Doc:

Which risk you mapped.

Which CIA principle it relates to.

Which configuration change you applied.

# Organize Information Following This Structure for Peer Review

## Title Page:

Project Title: “Applying the CIA Triad in Endpoint Security”

Your Name

Date

**Requirement 1:** Map three risks to CIA principles.

**Requirement 2:** Apply one Windows setting for each CIA principle.

**Requirement 3:** Document screenshots and explanations.

# Instructions for Documenting and Sharing the Project for Peer Review

## Document the Project



Use Word/Google Docs.

Insert screenshots of USB restriction, update settings, and service configuration.

Proofread for clarity.



## Share the Project

- Save as PDF.
- Upload to the “Peer Review” area.
- Provide a short summary in your post.



## Peer Review

- Review two peers’ submissions.
- Comment on whether they mapped CIA risks correctly and if their controls are effective.